

Systematic Review and Meta-Analysis on the Effect of Contrast Training on Vertical Jump Performance

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Abstract

The purpose of this study was to perform a systematic review and meta-analysis on the effect of contrast training (CT) on vertical jump performance. Thirteen out of 83 studies were included for systematic review while ten studies were utilized for meta-analysis. Meta-analysis revealed greater CMJ gains in CT than resistance training (ES = 1.30; CI = 0.31, 2.30). Similarly, CT displayed higher countermovement jump (CMJ) than control (ES = 1.46; CI = 0.46, 2.46). In conclusion, CT showed greater improvement in CMJ compared to RT and CON. For a video abstract of this article see Supplemental Digital Content 1, [HTTP://](http://LINKS.LWW.COM/SCJ/A247)

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Key Words contrast training, complex training, vertical jump, combined training, plyometric training, resistance training

INTRODUCTION

The ability to generate power plays a vital role in sport success (6,29). Power is defined as the capacity of the neuromuscular system to generate maximal force in the shortest period of time or at high velocity (13). The vertical jump (VJ) is a common task in sports performance that is dependent on lower body power. Strength and conditioning practitioners in sports with jumping tasks consider VJ as one of the critical parameters in monitoring strength and conditioning training adaptations. In the past decade, complex training has received notable attention as a training scheme to improve power (10,5,18,19,39). Complex training combines heavy resistance and plyometric/ballistic/speed training in one training session in an effort to enhance the force-velocity continuum characteristics of an individual. The heavy resistance training in complex training addresses the force curve component by increasing muscle force output while the high velocity training targets the velocity curve by producing force at high speeds or over a short period of time (5,13,31). The neurophysiological mechanism in complex training has been attributed to postactivation potentiation (PAP), which refers to an improvement in muscle kinetics from enhanced muscle phosphorylation via calcium sensitivity and h-reflex activity

(25,34,35). One complex training variation is contrast training (CT), which is performed by alternating a set of resistance exercise with a set of plyometrics or speed drill (4,15,31). This variation has been proposed to promote an acute ergogenic effect and enhance subsequent performance in power-oriented tasks (10,15).

Given the increasing interest in CT as a means to enhance performance, it is important to gain consensus on the literature on the topic to determine its efficacy and develop evidenced-based recommendations on its use. To date, no systematic review or meta-analysis has been carried out to accomplish this task. Thus, the purpose of this paper was to conduct a systematic review and meta-analysis on the effect of CT on vertical jump performance.

METHODS

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement was used to carry out a literature search in GoogleScholar, SPORTDiscus, World of Science, SpringerLink, and PubMed from all time points until January 30, 2018. The following search terms and Booleans were used: (complex training) OR (contrast training) OR (combined weight training and plyometrics) OR (combined strength training and plyometrics) OR (combined resistance training and plyometrics) AND (vertical jump or jump performance). Additionally, manual searches from relevant journals and reference lists from articles were conducted. To be included in the meta-analysis, studies had to meet the following criteria: 1) randomized trials peer-reviewed in English; 2) complex training intervention comparing any resistance training (RT) or plyometric training (PLYO) or control (CON), wherein complex training involves a set of resistance training followed by a set of high velocity exercise/plyometrics; 3) included

countermovement jump (CMJ) as a dependent variable; and, 4) training intervention performed at least twice a week with a duration of ≥ 4 weeks.

A single investigator (JP) initially assessed the eligibility of the studies for inclusion. In the first stage of screening, titles and abstracts of identified articles were checked for relevance. Reference lists of included articles were also checked for possible inclusion. In the second stage, full-text articles of potential studies were retrieved and assessed individually. Each study was evaluated for four times separated by 7 days to reduce selection bias. Studies that were deemed eligible were coded based on the following variables: author/s, participant information, description of exercise/s and activity, intensity of exercise/s, training frequency, study duration, CMJ test protocol, and study result. A second investigator (HP) independently checked the data extraction. After coding of data, both investigators rated the included studies for 'risk of bias' using an eight-point scale from Consolidated Standards of Reporting Trials (CONSORT) statement. The instrument uses a binary scale with each item scored as either 0 (absently or inadequately described) or 1 (explicitly described and present). A study with a score of 0-2 is regarded as having a high risk of bias, 3-5 with medium risk, and 6-8 considered as having low risk of bias. A consensus was reached for any disagreement presented in data extraction and CONSORT output.

Meta-analysis was carried out using a free software program (RevMan ver 5.3, The Nordic Cochrane Centre, Copenhagen). The meta-analysis aimed to examine CMJ height, considered as continuous variable, between: CT vs RT; CT vs PLYO; and, CT vs CON. The standard mean difference (mean CMJ difference from CT and comparison group/pooled standard deviation), also known as effect size (ES), was used to compare CMJ differences between groups (11,40). ES was interpreted using the following classification system: .2 – small effect; 0.5 0 moderate

effect; .8 – large effect. An inverse-variance random effects model was used due to the heterogeneous study methods and subject populations. Statistical heterogeneity was examined via chi-squared and I^2 -Index tests (24). An I^2 value greater than 25% is considered to exhibit low heterogeneity, 50% as moderate, and 75% as high heterogeneity.

A subgroup analysis was also carried out to identify potential moderating variables for any difference in CMJ between CT and comparison group. Variables included age (≥ 20 years vs. ≤ 20 years), level of physical ability (competitive vs. non-competitive). Training load intensity (≥ 70 RM vs. ≤ 70 RM) and study duration (≥ 6 weeks vs. ≤ 6 weeks) were also included in the subgroup analysis.

RESULTS

The literature search uncovered 1067 potential articles and two articles identified from reference lists. Removal of duplicates ($n = 345$) left 742 articles. After screening of title and abstracts, 83 articles underwent a more detailed evaluation and led to the exclusion of 70 articles. Thirteen studies (1,2,3,8,13,17,20,22,26,28,30,37,38) were ultimately deemed to meet the eligibility criteria for systematic review. Three studies from the systematic review (3,8,30) failed to qualify for meta-analysis due to missing CMJ data for pre-study, post-study, or both. This left ten studies (1,2,13,17,20,22,26,28,37,38) for inclusion in the meta-analysis. A flow of the search process is displayed in Figure 1.

Risk of bias of studies based on CONSORT is presented in Table 1 (insert table 1). Eleven studies (1,3,8,13,17,20,22,26,30,37,38) fall within the category of “medium risk” for bias while two studies (2,28) fall within the “high risk” category.

The total number of participants involved in the review were 441 (males, $n = 408$; females $n = 33$) with ages ranging from 8 to 30 years old. Nine studies (1,2,8,13,17,20,22,28,37) included an athlete population while 3 studies (26,30,38) employed an active but non-athletic population. Only one study was carried out in an untrained population (3). CT using pairing (1,3,13,22,30,37,38), triad (2,8,26) and combinations of both (17,20,28) were among CT strategies identified. A majority of studies (1,2,3,8,13,17,20,22,26,28,30,38) were carried out two days a week; only one study (34) employed a frequency of 3 times a week. Interventions lasted from 4 to 12 weeks. For identifying CMJ performance, nine studies utilized the hands-on-waist CMJ protocol (1,2,8,17,20,22,26,28,30). In one study the Vertec was employed to measure CMJ using the Jump and Reach protocol (13) and one study used the chalk method (38). One study also used a modified jump and reach protocol without arm swing (3). Lastly, CMJ in one study was measured using an unloaded bar in a Smith machine (37). Characteristics of studies are displayed in Table 2.

In the meta-analysis, four studies compared CT with RT (1,13,22,37), one differentiated CT and PLYO (13) and six studies compared CT with CON (1,17,20,22,28,37). The CON in one study was re-classified into RT because participants underwent stability and light resistance training (1).

Outcomes

Table 3 displays the CMJ output from CT, RT, PLYO and CON.

CT vs RT

Five out of ten studies compared CT with RT (1,13,22,26,37). These studies were found to have considerable heterogeneity at $\chi^2 = 21.03$, $df = 4$, $p = .0003$, $I^2 = 81\%$. Meta-analysis showed an

effect favoring CT compared to RT with an ES of 1.30 [0.31, 2.30], $Z = 2.56$, $p = 0.01$. CMJ from CT showed a 12.7% (95% CI 3.11 cm to 6.51 cm) increase compared to a 5.90% (95% CI 0.95 cm to 3.72 cm) CMJ enhancement in RT. Figure 2 depicts the forest plot comparing CT vs RT.

Subgroup analysis in CT vs. RT (Table 3) revealed a non-significant CMJ improvement for age (20 years $\geq$ vs. $\leq$ 20 years; $p = 0.08$) with CT compared to RT. However, longer participation in CT showed a significantly greater CMJ performance (≥ 6 weeks vs. ≤ 6 weeks; $p < .00001$) than RT.

CT vs CON

Comparison between CT and CON was analyzed in six out of 10 studies. These studies displayed high heterogeneity at $\chi^2 = 33.32$, $df = 5$, $p < 0.00001$, $I^2 = 85\%$. Meta-analysis showed greater CMJ improvements in CT compared to CON with an ES difference of 1.46 [0.46, 2.46], $Z = 2.85$, $p = 0.004$. On a percentage basis, CT resulted in an increase in CMJ of 8.6% (CI 95% 1.17 cm to 4.61 cm) whereas CON showed a decrement in performance of -0.91% (CI 95% -2.38 cm to 1.13 cm). Figure 4 depicts the forest plot comparing CT and CON. Subgroup analysis in CT vs. CON (Table 4) showed a non-significant difference in CMJ between for subjects 20 years $\geq$ and $\leq$ 20 years, $p = 0.44$. There was also no significant difference in CMJ between ≥ 70 RM and ≤ 70 RM loading schemes ($p = 0.34$). No subgroup difference in CT duration (≥ 6 weeks vs. ≤ 6 weeks; $p = 0.05$) was also observed.

DISCUSSION

The purpose of this study was to conduct a systematic review and meta-analysis on the effect of CT, a complex training strategy that involves completing a set of resistance exercise followed by a set of high velocity exercise, on VJ performance. In this review, CMJ was used as the lower

body power parameter when comparing CT with RT/PLYO/CON. Thirteen studies were included in the systematic review with CT interventions administered from healthy untrained population to elite-level athletes of various ages population. CT schemes ranged from the traditional complex pairing method to triad or a combination of both, carried out twice to thrice a week over a 4-12 week study period. In the meta-analysis, ten out of 13 studies were analysed. An analysis comparing CMJ output from CT and PLYO was discarded due to lack of available studies to qualify for meta-analysis (13). When comparing CT vs. RT, results showed that CT enhanced CMJ to a markedly greater extent compared to R ($ES = 1.30$; i.e., large effect). Similar findings were exhibited with CT in comparison with CON ($ES = 1.46$; i.e., large effect). Superior increases resulting from CT can be attributed to a PAP-induced phosphorylation and excitation of h-reflex (16,12,33,35). This hypothesis is partially supported by Labib (2013), who showed increased CD34/CD45 immune system stem cell secretions after CT (14,27,36). Moreover, evidence shows CT preserves type IIX muscle fibers, which are responsible for generating faster contraction velocities, power, and rate of tension development than type IIA and type I fibers (7,23,38). Another possible explanation for CT-induced enhancement in CMJ performance may be linked to greater volume in CT than other interventions, which may in turn have produced a greater stimulation of the neuromuscular system (41). However, it should be noted that higher training volumes may induce higher levels of neuromuscular fatigability than in the other groups (21,40). This, in turn, may negatively influence neuromuscular adaptations when combined with sport training (10,33,41).

In this study, subgroup analysis was conducted to determine possible moderators that may have contributed to the difference in CMJ between CT and RT. These covariates included age, training level, loading intensity for resistance exercise, and duration of intervention. Both subgroups for age improved CMJ; however, no difference in CMJ enhancement was seen between age groups. It should be noted that one study in the age subgroup was not included due

to ambiguous age data (13). Analysis for loading intensity was not possible due to absence of ≤ 70 RM group. Engaging in CT for both shorter (≤ 6 weeks) CT duration did not show improvement in CMJ compared to RT. On the other hand, there was a significant enhancement in CMJ with ≥ 6 weeks of CT than RT. There was a significant difference in CMJ between both CT duration. This may be linked to reduction of fatigue induced by CT at the onset of its intervention (14,25,40). No analysis was carried out as to level of activity due to the presence of only one study in the non-competitive group.

Subgroup analysis was also administered in CT vs. CON. Analysis showed that athletes in ≤ 20 -year old group exhibited greater gains in CMJ compared to athletes ≥ 20 years old; however, the difference between subgroups was non-significant. Also, the training load ≤ 70 RM showed a similar CMJ improvement with ≥ 70 RM, with no significant difference in CMJ scores noted between the two CT loading schemes. One study subgroup analysis for loading schemes was not included due to combining both loading schemes during the intervention (17). Implementation of CT for a longer duration (≥ 6 weeks) showed increased CMJ; however, no CMJ enhancement was found in CT duration of ≤ 6 weeks. The CMJ difference between the two duration subgroups was non-significant. Subgroup analysis of the level of physical activity was not possible due to limited available studies in the non-competitive subgroup.

There are several limitations to this meta-analysis that must be considered when attempting to draw practical inferences. First and foremost, analysis is based on a fairly small number of studies on the topic (10 total), and there was substantial heterogeneity in methodology between these studies. In particular, differences exist in the use of specific exercises (i.e. high pulls, squats, box jumps, etc.) as well as the CMJ testing measure (i.e. hands on waist, vertec, chalk, etc.). Thus, although the data do indicate a positive benefit to CT, the ability to draw strong inferences is limited. Moreover, the combination of mechanical stimuli within a CT

program may impair the ability to determine direct causality (19,41). The review also included CT in comparison with RT/PLYO/CON groups only. Future studies including other complex training and resistance training (e.g. compound training) schemes comparing CT is warranted. Further, only CMJ was utilized as the dependent variable; other performance parameters may produce different outcomes from CT and provide valuable information on the mechanical effects of CT on VJ. Lastly, the findings are limited to athletes and healthy non-athletes and cannot necessarily be generalized to other populations.

In conclusion, transference of CT in vertical jump enhancement was superior in comparison with RT and CON. However, changes in CMJ from the use of CT compared with PLYO are unclear.

PRACTICAL APPLICATION

Current evidence supports CT having greater efficacy in improving VJ performance when compared with RT and CON. For sports that include VJ actions, integration of CT for twice to thrice a week for more than 6 weeks into the comprehensive sports training program can help increase leg power in athletes. A common CT exercise scheme for the lower body involves a multijoint exercise (i.e. squat, leg press) followed by a VJ activity (i.e. CMJ jumps). However, coaches may use other CT variations by adding other plyometric/speed exercise(s) after the VJ task. In addition, a rest interval in between exercises (2-6 minutes) should be employed to maximize CT results. Lastly, when CT is carried out prior to sports training, applying one CT exercise performed for 3-5 sets and a 10-20 minute active recovery after CT would seem to be a sound strategy.

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Figure Legends

Figure 1: Flow diagram of search process

Figure 2: Forest plot comparing contrast training to resistance training

Figure 3: Forest plot comparing contrast training to control

ACCEPTED

Table 1									
CONSORT Scores of Studies Included in the Systematic Review									
Study	Were the groups comparable at baseline on key characteristics?	Did the study include a true control group (randomised participants – not a comparison group)	Was the randomisation procedure adequately described and carried out?	Did the study report a power calculation and was the study adequately powered to detect intervention effects?	Were the assessors blinded to treatment allocation at baseline and posttest?	Did at least 80% of participants complete follow up assessments?	Did the study analyses account for potential differences at baseline?	Did the study report effect sizes?	Total
Álvarez et al. 2012	1	0	0	0	0	1	1	1	4
Alves et al. 2010	0	0	0	0	0	1	1	0	2
Arazi et al. 2013	0	0	0	1	0	1	1	1	4
Brito et al. 2014	1	0	0	0	0	1	1	1	4
Dodd & Alvar 2010	0	0	0	0	0	1	1	1	3
Faude et al. 2013	0	0	0	0	0	1	1	1	3
García-Pinillos et al. 2014	0	0	0	1	0	1	1	0	3
Hammami et al. 2017	0	0	0	0	0	1	1	1	3
Juárez et al. 2009	0	0	1	0	0	1	1	0	3
Latorre Román et al., 2017	0	0	0	0	0	1	1	0	2
Macdonald et al. 2013	0	0	0	0	0	1	1	1	3
Spinetti et al. 2016	1	0	0	1	0	1	1	1	5
Stasinaki et al. 2015	1	0	0	0	0	1	1	1	4

Table 2							
Participants		Characteristics of Studies					
Authors	Participants	Intervention	Intensity	Frequency	Duration	Test parameter	Result
Mouret et al. 2012	male right handed male golfers	CT: horizontal bench press, seated row machine, barbell squats, seated military press;	3 x 5 @ 85% RM	2/7 days	12 weeks	hands-on-strap CMJ (cm)	CT > RT
	CT (n = 5; 24.2 ± 5.4 yrs), RT (n = 5; 25.9 ± 6.7 yrs)	seated calf-extension, triceps cable pushdown (Week 1 to 6)				on a jumping mat (Speed/UMF System, DSD, Spain)	
		horizontal bench press + plyo push-ups, seated row machine +	3 x 6 @ 70% RM + 3 x 10 reps for the				
		explosive pull-downs, barbell squat + vertical jumps over hurdles (45 cm);	first 3 complexes; last complex: 3 x 6 @ 70% RM +				
		seated barbell military press + plyo push-ups, seated calf-extension +	3 x 10 reps				
		vertical jumps over hurdles (45 cm); Triceps cable pushdown + plyometric					
		push-ups (Week 6 to 12); Day 1 and Day 2 extension of RT;					
		golf training: full round (18 holes)/week		5/7 days, 1-2/7 days	12 weeks		
		RT: core training (Day 1); light RT using possible rubber bands, ropes, or resistance from the trainer (Day 2);		2/7 days	12 weeks		
		front defolc raises, military press, abdominal crunches,					
		one- and two-arm keeps cards, standing backbends, sit-ups,					
		extensions, overhead sit-ups extensions, standing back rows,					
		lunges, squats, glute backbends, bent-over rows, hamstring					
		curls, and different forearm exercises					
		golf training: full round (18 holes)/week		5/7 days, 1-2/7 days	12 weeks		
Abou et al. 2010	male soccer players	CT: 1st complex: squat + high dips (5 m) + sprint (5 m); 2nd complex: standing calf raise + V1 + half heades;	1st complex: 1 x 6 @ 90% RM + 1 x 1 + 1 x 1	2/7 days	6 weeks	hands-on-strap CMJ (cm)	CT > CON
	CT (n = 8), CON (n = 6)	3rd complex: leg extension + VJ from seated position + Depth jump (60 cm box height) to heades	2nd complex: 1 x 6 @ 90% RM + 1 x 8 + 1 x 3 reps			on a jumping mat (ErgoJump, Globus Inc., Italy)	
		regular soccer training	3rd complex: 3 x 6 @ 80% RM + 6 reps 1 x 5		6 weeks		
		CON: regular soccer training			6 weeks		
Nani et al. 2014	healthy untrained women	CT: half squat + depth jumps; knee extension + multiple jumps;	resistance exercises: 3 x 6 @ 60% RM	2/7 days	6 weeks	CMJ without armwing (VERTTEK; Power Systems, Knoxville, Tennessee)	CT = RT = PLYO
	CT (n = 7; 20.8 ± 0.70 yrs), RT (n = 8; 20.7 ± 1.1 yrs)	knee flexion + zigzag jumps; single leg lunge + lunge jump	plyometrics exercises: 3 x 6 @ ME				
	PLYO (n = 8; 20.6 ± 1.10 yrs)	RT: half squat; knee extension; knee flexion; single leg lunge	3 x 6 @ 60% RM	2/7 days	6 weeks		
		PLYO: depth jump; multiple jumps; zigzag; dellics; lunge jump	3 x 6 @ ME	2/7 days	6 weeks		
Bauer et al. 2014	adult male soccer players	CT: squat + dips + sprint; calf extension on leg press device +	1 x 6 @ 85% RM + 5 m @ 90% ME + 5 m @ ME; 1 x 6 @ 90%	2/7 days	9 weeks	hands-on-strap CMJ using contact mat (DigiTime 1000; DigiTime, Finland)	CT = RT = PLYO
	CT (n = 12; 19.9 ± 0.50 yrs), RT (n = 12; 20.3 ± 0.90 yrs)	vertical jumps + half heades; leg extension + vertical jumps from	RM + 1 x 8 @ ME + 1 x 3 @ ME; 1 x 6 @ 80% RM +				
	PLYO (n = 12; 20.0 ± 0.60 yrs), CON (n = 21; 20.7 ± 1.00 yrs)	seated position + depth jumps (60 cm box height) to soccer heading	1 x 6 @ ME + 1 x 3 m 1 x 3 @ ME				
		regular soccer training	%5 incremental load from 1 RM every 3 weeks		9 weeks		
		RT: squat; calf extension; leg extension	1 x 6 @ 85% RM; 1 x 6 @ 90% RM; 1 x 6 @ 80% RM;	2/7 days	9 weeks		
		regular soccer training	%5 incremental load from 1 RM every 3 weeks				
		PLYO: dips + sprint; vertical jump + half heades; vertical jumps	1 x 5 m + 5 m @ ME; 1 x 8 + 1 x 3 @ ME; 1 x 3	2/7 days	9 weeks		
		from seated position + depth jumps (60 cm box height) to soccer heading; regular	0: 1 x 3 @ ME				
		regular soccer training			9 weeks		
Dodd et al. 2010	male junior college baseball players (18-23 yrs)	CT: squat + box jumps; lunge + depth jumps; split squat + split squat jump	2 x 6 @ 80-90% RM + 2 x 6 @ 0-30% RM	2/7 days	4 weeks	Jump and Reach using Vector (Jump USA, USA)	CT > RT
	CT (n = 32); RT (n = 31); PLYO (n = 28)	baseball specific skill and cardiovascular routines	45 mins	5/7 days	4 weeks		
		RT: squat; lunge; split squat	4 x 6 @ 80-90% RM	2/7 days	4 weeks		CT > PLYO
		baseball specific skill and cardiovascular routines	45 mins	5/7 days	4 weeks		
		PLYO: box jumps; depth jumps; split squat	4 x 6 @ 0-30% RM	2/7 days	4 weeks		
		baseball specific skill and cardiovascular routines	45 mins	5/7 days	4 weeks		

CT: control training; RT: resistance training group; PLYO: plyometric training group; CON: control group; RM: maximum repetition; PP: peak power; ME: maximal effort; CMJ: countermovement jump.

Authors	Participants	Exercises	Intervall	Frequency	Duration	Test parameters	Results
Pauole et al. 2013	high-level amateur football players CT (n = 8; 23.1 ± 2.7 yrs); CON (n = 8; 22.6 ± 2.4 yrs)	R.T. unilateral loaded half-squat + single-leg hurdle jump (Day 1); combination: two of the following exercises (Day 2): squat + drop jumps + 3 sprints loaded calf raises + straight jumps + header lateral half-squat + lateral jumps + zigzag sprints loaded step-ups + bounding jumps + header	4 x 4 @ 90% RM + 4 x 5	2/7 days	7 weeks	hands-on-wait CMJ using force platform (Kistler Type 9280AA, Winterthur, Switzerland)	CT > CON
		technical-tactical training	-	-	7 weeks		
		CON: technical-tactical training	-	-	7 weeks		
Guarda-Pedrol et al. 2014	male young soccer players CT (n = 17; 18.47 ± 1.26 yrs); CON (n = 13; 16.36 ± 1.13 yrs)	CT: isometric half-squat + jump from seated position (Week 1-2); isometric half-squat + single-leg squat, alternating right and left (Week 3-6); isometric half-squat + alternately (1) turn from seated position/ single-leg jump, alternating right and left (Week 3-6); isometric half-squat + single-leg jump, alternating right and left + isometric half-squat + jump from seated position (Week 7-8); Identical no weeks 7-8 (Week 9-10); Identical no weeks 11-12 (Week 11-12); normal soccer training CON: normal soccer training	4 x 40 secs + 4 x 5 4 x 60 secs + 3 x 4 6 x 40 secs + 6 x 4 4 x 40 secs + 4 x 6 + 4 x 40 secs + 4 x 6 5 x 40 secs + 5 x 6 + 5 x 40 secs + 5 x 6 6 x 40 secs + 6 x 6 + 6 x 40 secs + 6 x 6	2/7 days	12 weeks	hands-on-wait CMJ using accelerometer FlexPower Jump Assessment, Broomfield, Italy	CT > CON
		normal soccer training	-	-	12 weeks		
		CON: normal soccer training	-	-	12 weeks		
Hamamura et al. 2017	male young soccer players CT (n = 16; 16.0 ± 0.5 yrs); RT (n = 16; 16.2 ± 0.6 yrs); CON (n = 12; 16.8 ± 0.2)	CT: squat + CMJ (Week 1-4); squat + CMJ + CMJ to 15 m sprint (Week 4-8) soccer training RT: squat soccer training CON: soccer training	3.5 x 3.5 @ 70-90% RM + 5 x 3 3.5 x 3.5 @ 70-90% RM + 3.5 x 5 + 3 x 1 90 min emphasizing skill activities at various intensities, offensive and defensive strategies; 25-30 minutes of continuous play with brief interruptions from the coach 3.5 x 3.5 @ 70-90% 90 min emphasizing skill activities at various intensities, offensive and defensive strategies; 25-30 minutes of continuous play with brief interruptions from the coach 90 min emphasizing skill activities at various intensities, offensive and defensive strategies; 25-30 minutes of continuous play with brief interruptions from the coach	2/7 days 4-5/7 days 2/7 days 4-5/7 days 4-5/7 days 4-5/7 days	8 weeks 8 weeks 8 weeks 8 weeks 8 weeks 8 weeks	hands-on-wait CMJ on a force platform Kjumpo jump, version 1.04, Kistler Instrument AG, Winterthur, Switzerland)	CT > RT CT > CON
		normal soccer training	-	-	8 weeks		
		CON: normal soccer training	-	-	8 weeks		
Jones et al. 2009	active male undergraduate students CT (n = 8; 19.1 ± 1.36 yrs); RT (n = 8; 20.0 ± 1.93 yrs)	CT: week 1-2: back squat + vertical jump + sprint; week 3-4: back squat + hurdle jump + sprint RT: back squat	week 1-2: 2 x 6 @ 70% RM + 2 x 5 + 2 x 20 m week 3-4: 2 x 6 @ 70% RM + 2 x 5 + 2 x 20 m week 1-4: 4 x 6 @ 70% RM; week 2: 5 x 6 @ 70% RM	2/7 days 2/7 days	4 weeks 4 weeks	hands-on-wait CMJ (Ergojump Boostversion)	CT = RT

Authors	Participants	Exercises	Intensity	Frequency	Duration	Test parameter	Result
Lacort, Román et al., 2017	male and female children basketball athletes	CT: isometric half squat + depth jumps from seated position+ jump from seated position (Week 1-2)	40 secs + 2 x 10 + 3 x 10	2/7 days	10 weeks	hands-on-wrist CMI using an infrared sensor	CT > CON
		CT (n = 30, m = 25, f = 5), CON (n = 28, m = 25, f = 3)	isometric half squat + depth jumps from seated position+ jump from seated position (Week 3-4)	50 secs + 3 x 10 + 4 x 10		(Opto-Cast System, Microgate, Bolzano, Italy)	
		isometric half squat + depth jumps from seated position+ jump from seated position (Week 5-6)	60 secs + 5 x 10 + 5 x 10				
		isometric half squat + depth jumps from seated position+ jump from seated position (Week 7-8)	70 secs + 6 x 10 + 7 x 10				
		isometric half squat with partner seating on top of the thigh + depth jumps from seated position+ jump from seated position (Week 9-10)	70 secs + 6 x 10 + 7 x 10				
		basketball training/competitive game	-	3/7 days, at least 1/7 days	10 weeks		
		CON: basketball training/competitive game	-	3/7 days, at least 1/7 days	10 weeks		
Mackinnell et al., 2013	recreationally trained college men	CT: Smith machine squat + lateral jumps, Romanian deadlift + depth jumps	week 1-3, day 1: 3 x 5.6 @ 75-82% RM + 3 x 5.6	2/7 days	6 weeks	hands-on-wrist CMI on force platform (AMTI OR6-7, Advanced Mechanical Technology Inc., Watertown, MA)	CT > RT > PLYO
		CT (n = 10, 22.3 ± 3.24 yrs), RT (n = 13, 22.0 ± 3.66 yrs), PLYO (n = 11, 20.6 ± 3.36 yrs)	week 1-3, day 2: 3 x 5.6 @ 60-67% RM + 3 x 6.6				
		call-ense + box jumps	week 3-6, day 1: 3 x 3.4 @ 85-90% RM + 3 x 3.3				
			week 3-6, day 2: 3 x 3.4 @ 45-55% RM + 3 x 3.4				
		RT: Smith machine squat, Romanian deadlift, call-ense	week 1-3, day 1: 5.6 x 5.6 @ 75-82% RM, week 1-3, day 2: 3 x 5.6 @ 60-67% RM, week 3-6, day 1: 3 x 3.4 @ 85-90% RM, week 3-6, day 2: 3 x 3.4 @ 45-55% RM	2/7 days	6 weeks		
		PLYO: lateral jumps, depth jumps, box jumps	week 1-3, day 1: 3 x 5.6, week 1-3, day 2: 3 x 4.6, week 3-6, day 1: 3 x 3.3, week 3-6, day 2: 3 x 3.4	2/7 days	6 weeks		
Spinelli et al., 2015	male soccer players (n = 22, 18.4 ± 0.4 yrs)	CT: CMI + frontal jumps over barrier (80 cm high, 80 cm long), CMI + single frontal jumps over barrier (50 cm high, 80 cm long box)	1 x 6 @ 80 PP + 1 x 10 x 1 x 6 @ 80 PP + 1 x 10	3/7 days	12 weeks	CMI with unbraked bar on Smith machine using accelerometer (Microfit, SA, Sion, Switzerland)	CT > RT
		CT (n = 22), RT (n = 10)	1 x 6 @ 80 PP + 1 x 10				
		CMI + multiple diagonal cone hops (40 cm high, 80 cm long box) (Day 1)	1 x 6 @ 80 PP + 1 x 10				
		high pull power + frontal sprint, high pull power + knee up to frontal sprint, high pull power + 110 degree curve zigzag sprints (Day 2)	1 x 8 RM + 1 x 10 m, 1 x 3 RM + 1 x 5 m + 1 x 10 m, 1 x 5 RM + 4 x 5 m				
		CMI + single jump on the box (50 cm box height), CMI + depth jumps ("Box onto box"), 80 cm box height), CMI + depth jumps (50 cm box height) (Day 3)	1 x 4 @ 100 PP, 1 x 1 x 10, 1 x 4 reps @ 100 PP + 1 x 10, 1 x 4 reps @ 100 PP, 1 x 10, 1 x 4 reps @ 100 PP, 1 x 10				
		technical skills and tactical training	-	-	12 weeks		
		RT: squat on Smith machine, deadlift knee flexion with olympic bar, stiff legged deadlift with olympic bar, knee extension machine, knee flexion machine, hip adduction machine (Day 1)	2 x 12-15 RM	3/7 days	12 weeks		
		squat on Smith machine, deadlift knee flexion with olympic bar, stiff legged deadlift with olympic bar, knee extension machine (Day 2)	3 x 8-10 RM				
		squat on Smith machine, deadlift knee flexion with olympic bar, stiff legged deadlift with olympic bar (Day 3)	4 x 4-6 RM				
		technical skills and tactical training	-	-	12 weeks		
Stansfeld et al., 2013	moderately-trained physical education students	CT: leg press + leg press throw, bench press + bench press throw, Smith squat + squat jump + drop jumps (every other training day)	2 x 6 @ 85% RM for heavy load + 2 x 8 @ 30% RM for the 2nd complex pair exercise	3/7 days	6 weeks	CMI using jump and Reach (chalk method)	CT > CON
		CT (n = 9), CON (n = 7)	3 x 8 @ 30-45 cm (drop jumps)				
		athletic throwing skills training	light intensity training for 2 hours	2/7 days	6 weeks		
		CON: athletic throwing skills training	light intensity training for 2 hours	2/7 days	6 weeks		

Table 3					
Subgroup Analysis for CT vs. RT					
Group	Studies				Subgroup difference
	No. (reference)	ES (95% CI)	I ²	p-value	p-value
<i>Population Characteristics</i>					
Age					
≥ 20 years	2 (1,26)	1.14(0.28, 1.99)	0	.009	0.08
< 20 years	2 (22,37)	2.19(1.41, 2.97)	0	< .00001	
Level					
competitive	5 (1,13, 22,37)	0.73(0.33, 1.15)	85	.0004	NA
non-competitive	1 (26)	1.11(0.04, 2.19)	NA	0.04	
<i>CT Training Characteristics</i>					
Loading Intensity					
≥ 70 RM	3 (1,13,22,26,37)	0.78(0.40, 1.15)	81	< .00001	NA
< 70 RM	NA	NA	NA	NA	
Duration					
≥ 6 weeks	3 (1,22,37)	1.95(1.26, 2.63)	0	< .00001	< .00001
< 6 weeks	2 (13,26)	0.43(-0.02, 0.87)	65	0.24	

Table 4					
Subgroup Analysis for CT vs. CON					
Group	Studies				Subgroup difference
	No. (reference)	ES(95% CI)	I ²	p-value	p-value
<i>Population Characteristics</i>					
Age					
≥ 20 years	2 (17,38)	0.72(-0.08, 1.51)	88	0.08	0.44
< 20 years	4 (2,20,22,28)	1.07(0.67, 1.46)	88	< .00001	
Level					
competitive	5 (2,17, 20,22,28)	1.17(0.79, 1.55)	85	< .00001	NA
non-competitive	1 (38)	-0.15(-1.14, 0.84)	NA	0.77	
<i>CT Training Characteristics</i>					
Loading Intensity					
≥ 70 RM	3 (2,22,38)	1.18(0.50, 1.85)	93	.0006	0.34
< 70 RM	3 (20,28)	0.79(0.35, 1.22)	0	.0004	
Duration					
≥ 6 weeks	3 (17,20,22,28)	1.19(0.79, 1.59)	89	< .00001	0.05
< 6 weeks	2 (2,38)	0.33(-0.41, 1.08)	53	0.38	





